

- Keep the thumbs crossed and hands close to—but not touching—the bar as it ascends.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.
- At the end of the set, move sideways in unison with the athlete back to the rack.
- Simultaneously grasp the bar and assist with balancing the bar as it is placed back on the supports.
- Release the bar smoothly.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

- **Create a shelf with your shoulders**
- **Keep your elbows high**
- **Sit between your heels**

COMMON ERRORS

- **Allowing the arms to relax or the elbows to drop down and backward**
- **Excessive forward lean or raising with the hips first**



Upward movement positions



Racking the bar

▶ 24 ZERCHER SQUAT

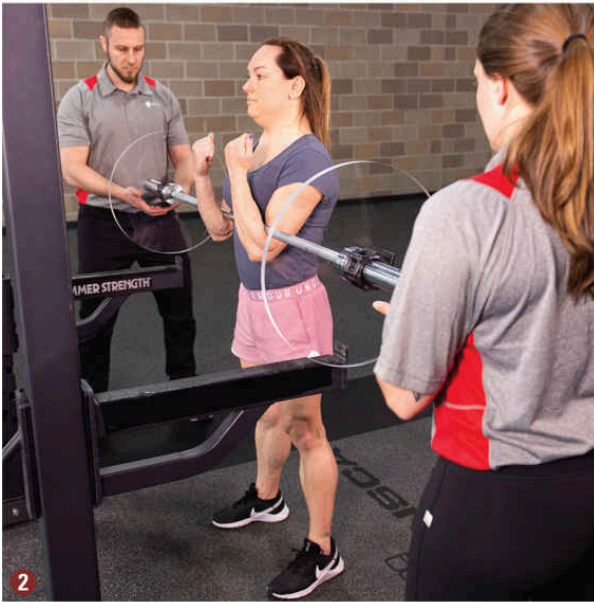
Starting Position: Athlete

- Set a barbell at approximately mid torso height on a squat rack or power rack.

- Approach the bar and position it in the crease of the elbows, resting on the biceps and held tight to the body.
- Step under the bar and position the feet parallel to each other.
- Engage the core, retract the shoulder blades, and hold the chest up and out.
- Tilt the head slightly up.
- Once in position, signal the spotters for assistance in moving the bar off the supports.
- Extend the hips and knees to lift the bar.
- Take one or two steps backward.
- Position the feet shoulder-width apart (or wider), even with each other, with the toes pointed slightly outward.
- All repetitions begin from this position.



1 Starting position on the rack



2 Starting position in the rack



3 Starting position out of the rack

Starting Position: Two Spotters

- Stand erect at opposite ends of the bar with the feet shoulder-width apart and the knees slightly flexed.
- Grasp the end of the bar by cupping the hands under the bar with the fingers across each other.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved off the supports.

- Release the bar smoothly.
- Hold the hands 2 to 3 inches (5-8 cm) below the ends of the bar.
- Move sideways in unison with the athlete as the athlete moves backward.
- Once the athlete is in position, get into a shoulder-width stance with the knees slightly flexed and the torso erect.

Downward Movement Phase: Athlete

- Maintain a position with the back neutral, the arms tight to the body, and the chest up and out.
- Allow the hips and knees to flex slowly while keeping the torso-to-floor angle relatively constant.
- Keep the heels on the floor and the knees aligned over the feet.
- Continue flexing the hips and knees until the tops of the thighs are parallel to the floor, the trunk begins to round or flex forward, or the heels rise off the floor.

Downward Movement Phase: Two Spotters

- Keep the hands cupped and hands close to—but not touching—the bar as it descends.
- Slightly flex the knees, hips, and torso to keep a neutral spine position when following the bar.

Upward Movement Phase: Athlete

- Maintain a position with a neutral spine, the elbows held to body, and the chest up and out.
- Extend the hips and knees at the same rate (to keep the torso-to-floor angle constant).
- Keep the heels on the floor and the knees aligned over the feet.
- Do not flex the torso forward or round the back.
- Continue extending the hips and knees to reach the starting position.
- At the end of the set, step forward toward the rack.

- Squat the bar down until it rests on the supports.

Upward Movement Phase: Two Spotters

- Keep the hands cupped and close to—but not touching—the bar as it ascends.
- Slightly extend the knees, hips, and torso, and keep the back neutral when following the bar.
- At the end of the set, move sideways in unison with the athlete back to the rack.
- Simultaneously grasp the bar and assist with balancing the bar as it is racked.
- Release the bar smoothly.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, biceps brachii

VERBAL CUES

- **Cradle the bar in your elbows**
- **Keep your elbows in**
- **Control your descent**

COMMON ERRORS

- **Not keeping the elbows close to the body**
- **Leaning too far forward**



Downward movement positions

▶ **25 FORWARD STEP LUNGE**

This exercise can also be performed with two dumbbells, using a closed, neutral grip. The athlete allows the dumbbells to hang at arm's length alongside the body throughout the exercise. When using dumbbells, the spotter assists the athlete in the same manner as in the barbell version, although the task of helping the athlete rack the bar is not applicable. (*Note:* This exercise cannot be performed inside a typical power rack. The use of a tiered squat rack or a power rack that has supports on the outside is recommended. To allow an optimal view of the exercise technique, a rack is not shown in the photos.)

Starting Position: Athlete

- Step under the bar and position the feet parallel to each other.
- Grasp the bar with a closed, pronated grip.
- Place the bar in a balanced position on the upper back and shoulders above the posterior deltoids at the base of the neck (using a handgrip only slightly wider than shoulder width).

- Lift the elbows up to create a shelf for the bar using the upper back and shoulder muscles.
- Hold the chest up and out.
- Tilt the head slightly up.
- Once in position, signal the spotter for assistance in moving the bar off the supports.
- Extend the hips and knees to lift the bar.
- Take two or three steps backward.
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect and very close to the athlete (but do not distract the athlete).
- Place the feet shoulder-width apart with the knees slightly flexed.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved out of the rack.
- Move in unison with the athlete as the athlete moves backward to the starting position.
- Once the athlete is in position, get into a hip-width stance with the knees slightly flexed and the torso erect.
- Stay in position in case the athlete needs assistance. Position the hands near the athlete's hips, waist, or torso if needed to help stabilize the athlete.

Forward Movement Phase: Athlete

- Take one exaggerated step directly forward with one leg (the lead leg).
- Keep the torso erect as the lead foot moves forward and contacts the floor.
- Keep the trailing foot in the starting position, but allow the trailing knee to slightly flex.
- Plant the lead foot flat on the floor pointing straight ahead or slightly inward.

- Allow the lead hip and knee to flex slowly.
- Keep the lead knee directly over the lead foot.
- Continue to flex the trailing knee until it is 1 to 2 inches (3-5 cm) above the floor.
- Balance the weight evenly between the ball of the trailing foot and the whole lead foot.
- Keep the torso perpendicular to the floor by sitting back on the trailing leg.



Starting positions



Beginning of forward movement positions

Forward Movement Phase: Spotter

- Stay in a hip-width stance with the knees slightly flexed and the torso erect.
- Assist only if necessary to keep the athlete balanced.

Backward Movement Phase: Athlete

- Forcefully push off the floor by extending the lead hip and knee.

- Maintain the same erect torso position; do not jerk the upper body backward.
- Bring the lead foot back to a position next to the trailing foot; do not stutter-step backward.
- Stand erect in the starting position, pause, and then alternate lead legs for the next repetition.
- At the end of the set, step toward the rack and place the bar in the supports.

Backward Movement Phase: Spotter

- Maintain a hip-width stance with the knees slightly flexed and the torso erect.
- Stand erect in the starting position, and pause to wait for the athlete to switch legs for the next repetition.
- Stay in a ready position with hips and knees slightly flexed, and assist only if necessary to keep the athlete balanced.
- At the end of the set, help the athlete rack the bar.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, iliopsoas

VERBAL CUES

- **Step forward leading with your heel**
- **Front shin vertical at the bottom**
- **Drive through your front heel**

COMMON ERRORS

- **Stepping out too shallowly, causing the lead knee to extend too far past the lead foot**
- **Allowing the torso to flex forward during the forward movement phase**



3 Completion of forward movement positions



4 Middle of backward movement positions

▶ **26** **REVERSE STEP LUNGE IN FRONT RACK POSITION**

Note: This exercise cannot be performed inside a typical power rack. The use of a tiered squat rack or a power rack that has supports on the outside is recommended. To allow an optimal view of the exercise technique, a rack and spotters are not shown in the photos.

Starting Position: Athlete

- Step toward the bar and position the feet parallel to each other.
- Place the hands on the bar in one of two arm positions:
 1. Parallel-arm position
 - » Grasp the bar with a closed, pronated grip.
 - » Grip should be slightly wider than shoulder width.
 - » Move up to the bar to place it on top of the anterior deltoids and clavicles.
 - » Fully flex the elbows to position the upper arms parallel to the floor.

2. Crossed-arm position

- » Flex the elbows and cross the arms in front of the chest.
- » Move up to the bar to place it on top of the anterior deltoids.
- » Use an open grip with the hands on top of the bar and the fingers holding it in place.
- » Lift the elbows to position the arms parallel to the floor.
- Hold the chest up and out.
- Tilt the head slightly up.
- Extend the hips and knees to lift the bar.
- Take one or two steps backward.
- Position the feet shoulder-width apart (or wider), even with each other, with the toes pointed forward.
- All repetitions begin from this position (*note*: this preparatory position is not shown).

Starting Position: Spotter

- Stand erect and very close to the athlete (but do not distract the athlete).
- Place the feet shoulder-width apart with the knees slightly flexed.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved out of the rack.
- Move in unison with the athlete as the athlete moves backward to the starting position.
- Once the athlete is in position, get into a hip-width stance with the knees slightly flexed and the torso erect.
- Stay in position in case the athlete needs assistance.



Backward (reverse) step

Backward Movement Phase: Athlete

- Take one exaggerated step directly backward with one leg.
- Keep the torso erect as the rear foot moves backward and contacts the floor.
- Keep the front foot in the starting position, but allow the backward knee to flex toward the floor.
- Keep the lead foot flat on the floor pointing straight ahead.
- Allow the front hip and knee to flex slowly.
- Keep the front knee directly over the front foot.
- Continue to flex the front knee until the rear knee is 1 to 2 inches (3-5 cm) above the floor.
- Balance the weight evenly between the ball of the trailing foot and the whole lead foot.

- Keep the torso perpendicular to the floor by sitting back on the rear leg.

Backward Movement Phase: Spotter

- Stay in a hip-width stance with the knees slightly flexed and the torso erect.
- Assist only if necessary to keep the athlete balanced.

Forward Movement Phase: Athlete

- Forcefully push off the floor by extending the front hip and knee.
- Maintain the same erect torso position; do not jerk the upper body forward.
- Bring the rear foot back to a position next to the lead foot; do not stutter-step forward.
- Stand erect in the starting position, pause, and then alternate lead legs for the next repetition.
- At the end of the set, step toward the rack and place the bar in the supports.

Forward Movement Phase: Spotter

- Maintain a hip-width stance with the knees slightly flexed and the torso erect.
- Stand erect in the starting position and pause to wait for the athlete to switch legs for the next repetition.
- Stay in a ready position with hips and knees slightly flexed, and assist only if necessary to keep the athlete balanced.
- At the end of the set, help the athlete rack the bar.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, iliopsoas, soleus, gastrocnemius

VERBAL CUES

- **Elbows high and forward**
- **Step back and lower straight down**
- **Drive through your front foot**

COMMON ERRORS

- **Leaning forward excessively**
- **Overextending the backward step**



Lowest lunge position

27 STEP-UP

Note: The box used should be 12 to 18 inches (30-46 cm) high, or high enough to create a 90-degree angle at the knee joint when the foot is on the box. To allow an optimal view of the exercise technique, a rack is not shown in the photos.

Starting Position: Athlete

- Grasp the bar with a closed, pronated grip.

- Step under the bar and position the feet parallel to each other.
- Place the bar in a balanced position on the upper back and shoulders above the posterior deltoids at the base of the neck (using a handgrip only slightly wider than shoulder width).
- Lift the elbows up to create a shelf for the bar using the upper back and shoulder muscles.
- Hold the chest up and out.
- Tilt the head slightly up.
- Once in position, signal the spotter for assistance in moving the bar off the supports.
- Extend the hips and knees to lift the bar.
- Move to a spot near the front of the box.
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect and very close to the athlete (but do not distract the athlete).
- Place the feet shoulder-width apart with the knees slightly flexed.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved out of the rack.
- Move in unison with the athlete as the athlete moves to the starting position.
- Once the athlete is in position, get into a hip-width stance with the knees slightly flexed and the torso erect.
- Stay in a ready position with hips and knees slightly flexed, and assist only if necessary to keep the athlete balanced.

Upward Movement Phase: Athlete

- Step up with one leg (the lead leg) to place the entire foot on the top of the box.
- Keep the torso erect; do not lean forward.

- Keep the trailing foot in the starting position, but shift the body weight to the lead leg.
- Forcefully extend the lead hip and knee to move the body to a standing position on top of the box.
- Do not push off or hop up with the trailing leg or foot.
- At the highest position, stand erect and pause before beginning the downward movement phase.

Upward Movement Phase: Spotter

- Take a small step forward with the lead leg as the athlete steps up on the box.
- As the athlete reaches the highest position, bring the trailing leg forward to be next to the lead leg.
- Stay in a ready position with hips and knees flexed, and assist only when necessary to keep the athlete balanced.



Starting positions



Initial contact of lead foot with top of box

Downward Movement Phase: Athlete

- Shift the body weight to the same lead leg.

- Step off the box with the same trailing leg.
- Maintain an erect torso position.
- Place the trailing foot on the floor the same distance from the box as the starting position.
- When the trailing foot is in full contact with the floor, shift the body weight to the trailing leg.
- Step off the box with the lead leg.
- Bring the lead foot back to a position next to the trailing foot.
- Stand erect in the starting position, pause, and then alternate lead legs for the next repetition.
- At the end of the set, step toward the rack and place the bar in the supports.

Downward Movement Phase: Spotter

- Take a small step backward with the same trailing leg as the athlete steps back down to the floor.
- As the athlete steps off the box with the lead leg, take a step backward with the same lead leg.
- Keep the hands near the athlete's hips, waist, or torso.
- Stand erect in the starting position and pause to wait for the athlete.
- Assist only if necessary to keep the athlete balanced.
- At the end of the set, help the athlete rack the bar.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

- **Chest up and shoulders back**
- **Drive through your heel, and push the box away from you**

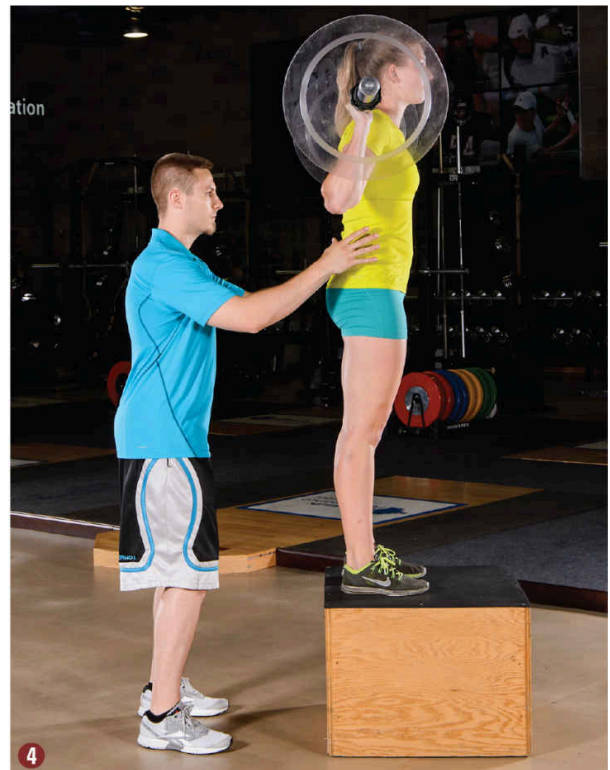
- **Stand tall at the top of the box**

COMMON ERRORS

- **Pushing off with the back leg to assist with upward movement**
- **Flexing too far forward at the hips, causing a forward lean**



Middle of upward movement positions



Completion of upward movement positions

28 GOOD MORNING

Note: To allow an optimal view of the exercise technique, a rack is not shown in the photos.

Starting Position

- Grasp the bar with a closed, pronated grip.
- Step under the bar and position the feet parallel to each other.

- Place the bar in a balanced position on the upper back and shoulders above the posterior deltoids at the base of the neck (using a handgrip only slightly wider than shoulder width).
- Lift the elbows up to create a shelf for the bar using the upper back and shoulder muscles.
- Hold the chest up and out.
- Tilt the head slightly up.
- Position the feet shoulder-width apart (or wider), even with each other, with the toes pointed slightly outward.
- Extend the hips and knees to lift the bar off the supports.
- Take two or three steps backward.
- All repetitions begin from this position.

Downward Movement Phase

- Begin the exercise by slowly permitting the hips to flex. The buttocks should move straight back during the descent.
- Maintain a neutral spine and high elbow position; do not round the upper back during the descent.
- Keep the knees slightly flexed and the feet flat on the floor during the descent.
- Continue the downward movement until the torso is approximately parallel to the floor.

Upward Movement Phase

- Raise the bar by extending the hips.
- Keep the back neutral and the knees slightly flexed during the ascent.
- Continue extending the hips to reach the starting position.
- At the end of the set, step toward the rack and place the bar in the supports.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, erector spinae

VERBAL CUES

- Soft flex of the knees
- Hinge at the hips and push your hips backward
- Chest up, shoulders back

COMMON ERRORS

- Rounding the back and not maintaining a neutral spine position
- Flexing the knees too much, turning the movement into a squat instead of a hip hinge



Starting position



Downward and upward movements

29 DEADLIFT

Starting Position

- Stand with the feet flat and placed between hip- and shoulder-width apart with the toes pointed slightly outward.
- Squat down with the hips lower than the shoulders (farther than shown in the first photo and video clip), and grasp the bar with a closed, pronated grip. If the load is too heavy to hold the bar with a pronated grip, change to a closed, alternated grip.
- Place the hands on the bar slightly wider than shoulder-width apart, outside of the knees, with the elbows fully extended.
- Place the feet flat on the floor and position the bar approximately 1 inch (3 cm) in front of the shins and over the balls of the feet.
- Position the body with the
 - back neutral or slightly arched,
 - scapulae depressed and retracted,
 - chest held up and out,
 - head in line with the vertebral column or slightly hyperextended,
 - heels in contact with the floor,
 - shoulders over or slightly in front of the bar, and
 - eyes focused straight ahead or slightly upward.
- All repetitions begin from this position.

Upward Movement Phase

- Lift the bar off the floor by extending the hips and knees.
- Keep the torso-to-floor angle constant; do not let the hips rise before the shoulders.
- Maintain a neutral spine position.
- Keep the elbows fully extended and the shoulders over or slightly ahead of the bar.
- As the bar is raised, keep it as close to the shins as possible.
- As the bar rises just above the knees, keep the shoulders over the bar and extend the hips to keep the bar close to the body.

- Continue to extend the hips and knees until the body reaches a fully erect torso position.

Downward Movement Phase

- Allow the hips and knees to flex to slowly lower the bar to the floor.
- Maintain the neutral spine position; do not flex the torso forward.

MAJOR MUSCLES INVOLVED

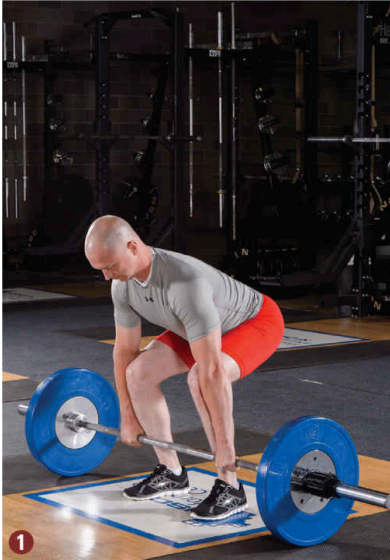
gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

- **Chest up and shoulders back**
- **Push the floor away from you**
- **Stand tall and squeeze your glutes**

COMMON ERRORS

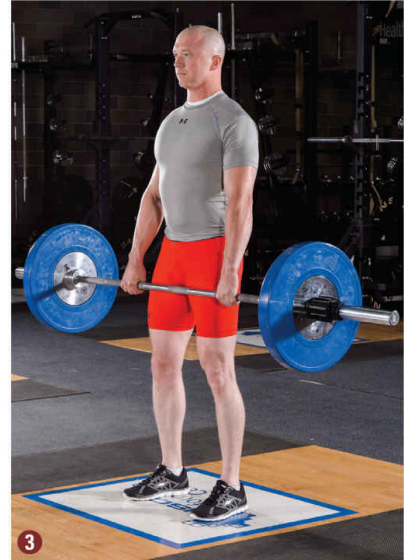
- **Allowing the bar to drift away from the body**
- **Failing to lock out completely**



1 Starting position



2 Middle position



3 End position

30 STIFF-LEG DEADLIFT

Starting Position

- After performing the deadlift exercise to lift the bar off the floor, slightly to moderately flex the knees and keep them in this position throughout this exercise.
- All repetitions begin from this position.

Downward Movement Phase

- Begin the exercise by forming a neutral spine, then flex at the hips to lower the bar slowly, and under full control, toward the floor.
- Keep the knees in the same slightly or moderately flexed position with the back neutral or slightly arched and the elbows fully extended during the descent.
- Lower the bar until the plates touch the floor; the back cannot move out of the neutral position, the knees fully extend, or the heels rise off of the floor.

Upward Movement Phase

- Extend at the hips to return to the standing starting position.

- Keep the knees slightly flexed and the torso in a neutral spine position.
- Do not jerk the torso backward or flex the elbows.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, erector spinae

VERBAL CUES

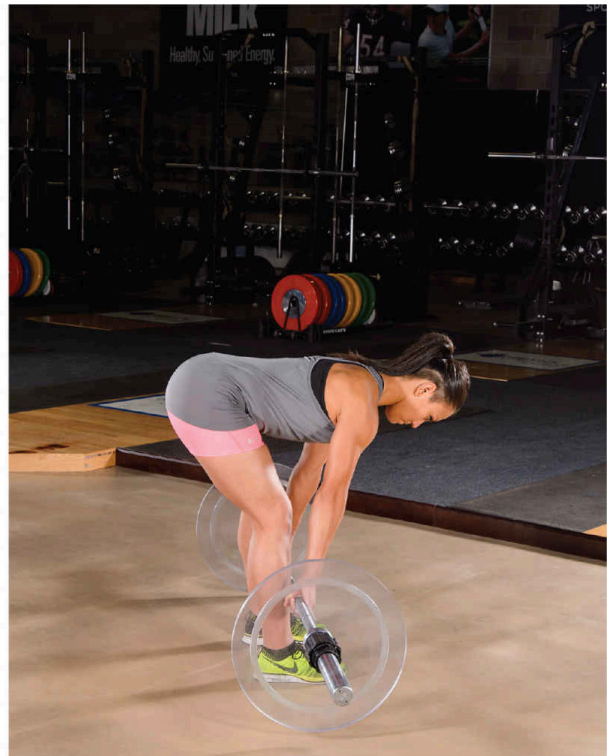
- Hinge at the hip
- Keep your back flat
- Keep your weight in your heels

COMMON ERRORS

- Not maintaining a neutral spine
- Bouncing the weight off floor
- Actively flexing the knees



Starting position



Downward and upward movements

31 ROMANIAN DEADLIFT (RDL) (AND SNATCH GRIP VARIATION)

Starting Position

- Place the hands on the bar in a closed, pronated position using either a clean or snatch grip.
- After performing the deadlift exercise to lift the bar off the floor, slightly to moderately flex the knees and keep them in this position throughout this exercise. (*Note:* The knees should be slightly more flexed than shown in the video.)
- All repetitions begin from this position.

Downward Movement Phase

- Begin the exercise by flexing the hips and pushing them backward, allowing the torso to move forward, keeping the bar in contact with the thighs.
- Keep the knees slightly flexed as the hips flex.
- Maintain a rigid torso, neutral spine, and keep the shoulders retracted until the barbell is aligned with the patella tendon and the torso is parallel to floor. (*Note:* If using a snatch grip with this exercise, the torso will be slightly below parallel, depending on the athlete's anthropometrics.)
- Keep a normal lordotic position throughout the movement.

Upward Movement Phase

- Extend the hips, raising the torso back to the starting standing position.
- Keep the knees slightly flexed and the torso in a neutral spine position.
- Make sure the barbell maintains contact with the thighs throughout the movement.
- Do not hyperextend the back or flex the elbows.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, erector spinae

VERBAL CUES

- **Soft flex of the knees**
- **Push your hips back**
- **Slide the bar down your thighs**

COMMON ERRORS

- **Rounding the back and not maintaining a neutral spine position**
- **Allowing the bar to drift away from the body**
- **Not pushing the hips back far enough, reducing hamstring engagement**



Starting position



Downward and upward movements

▶ **32** **LEG (KNEE) EXTENSION** **(MACHINE)**

Starting Position

- Sit down in the machine and press the back firmly against the back pad.
- Place the feet behind and in contact with the roller pad.
- Position the legs parallel to each other.

- Align the knees with the axis of the machine. If necessary, adjust the back pad or the roller pad to position the legs correctly.
- Grasp the handles or the sides of the seat.
- All repetitions begin from this position.

Upward Movement Phase

- Raise the roller pad by fully extending the knees.
- Keep the torso erect and the back firmly pressed against the back pad.
- Keep the thighs, lower legs, and feet parallel to each other.
- Maintain a tight grip on the handles or the sides of the seat.
- Do not forcefully lock out the knees.

Downward Movement Phase

- Allow the knees to slowly flex back to the starting position.
- Keep the torso erect and the back firmly pressed against the back pad.
- Keep the thighs, lower legs, and feet parallel to each other.
- Do not allow the buttocks to lift off the seat.
- Maintain a tight grip on the handles or the sides of the seat.

MAJOR MUSCLES INVOLVED

vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

- **Point your toes toward your shins**
- **Lift with your thighs**
- **Slow and controlled on the way down**

COMMON ERRORS

- **Allowing the hips or buttocks to lift off the seat during the upward movement**

- **Swinging or jerking the torso backwards to help raise the weight**
- **Forcefully locking out the knees at the end of the upward movement phase**



Starting position



Upward and downward movements

▶ 33 **SEATED LEG (KNEE) CURL** **(MACHINE)**

Starting Position

- Sit down in the machine and press the back firmly against the back pad.
- Place the ankles on top of and in contact with the roller pad.
- Position the legs parallel to each other.
- Align the knees with the axis of the machine. If necessary, adjust the back pad to position the legs correctly.
- Grasp the handles or the sides of the seat.
- All repetitions begin from this position.

Downward Movement Phase

- Curl the roller pad below the seat by fully flexing the knees.
- Keep the torso stationary and the hips and torso firmly pressed against the pads.
- Do not allow the hips or low back to lift off the pads.
- Maintain a tight grip on the handles or the sides of the seat.

Upward Movement Phase

- Allow the knees to slowly extend back to the starting position.
- Keep the torso stationary and the hips and low back firmly pressed against the pads.
- Maintain a tight grip on the handles or the sides of the seat.
- Do not forcefully lock out the knees.

MAJOR MUSCLES INVOLVED

semimembranosus, semitendinosus, biceps femoris

VERBAL CUES

- **Pull your heels toward your glutes**
- **Maintain tension throughout the movement**
- **Keep your upper body still**

COMMON ERRORS

- **Raising the hips off the seat to assist in the movement**
- **Allowing the weight to fall back too quickly during the eccentric phase**
- **Incomplete ROM**



Starting position



Downward and upward movements

Shoulders

34 SHOULDER PRESS (MACHINE)

Starting Position

- Sit down and lean back to place the body in the five-point body contact position.
- Grasp the handles with a closed, pronated grip.
- Align the handles with the top of the shoulders. If necessary, adjust the seat height to position the handles correctly.

Upward Movement Phase

- Push the handles upward until the elbows are fully extended.
- Maintain the five-point body contact position.
- Do not arch the lower back or forcefully lock out the elbows.

Downward Movement Phase

- Allow the elbows to slowly flex to lower the handles to the starting position.
- Maintain the five-point body contact position.

MAJOR MUSCLES INVOLVED

anterior and medial deltoids, triceps brachii

VERBAL CUES

- **Sit tall with your feet flat on the floor**
- **Push your hands toward the ceiling**
- **Control the descent**

COMMON ERRORS

- **Excessively arching the back during the upward movement phase**
- **Not executing the movement through a full ROM**



Starting position



Upward and downward movements

35 SEATED BARBELL SHOULDER PRESS (AND DUMBBELL VARIATION)

This exercise can also be performed with two dumbbells, using a closed, pronated grip. When using dumbbells, the spotter assists by spotting the athlete's forearms near the wrists instead of the bar.

Starting Position: Athlete

- Sit down on a vertical shoulder press bench and lean back to place the body in the five-point body contact position.
- Grasp the bar with a closed, pronated grip.
- Grip should be slightly wider than shoulder width.
- Signal the spotter for assistance in moving the bar off the supports.
- Press the bar over the head until the elbows are fully extended.
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect behind the bench with the feet shoulder-width apart and the knees slightly flexed.
- Grasp the bar with a closed, alternated grip inside the athlete's hands.
- At the athlete's signal, assist with moving the bar off the supports.
- Guide the bar to a position over the athlete's head.
- Release the bar smoothly.

Downward Movement Phase: Athlete

- Allow the elbows to slowly flex to lower the bar.
- Keep the wrists stiff and the forearms parallel to each other.
- Extend the neck slightly to allow the bar to pass by the face as it is lowered to touch the clavicles and anterior deltoids.
- Maintain the five-point body contact position.

Downward Movement Phase: Spotter

- Keep the hands in the alternated grip position close to—but not touching—the bar as it descends.
- Keep the knees slightly flexed and the back neutral when following the bar.

Upward Movement Phase: Athlete

- Push the bar upward until the elbows are fully extended.
- Extend the neck slightly to allow the bar to pass by the face as it is raised.
- Keep the wrists stiff and the forearms parallel to each other.
- Maintain the five-point body contact position.
- Do not arch the back or rise off the seat.
- At the end of the set, signal the spotter for assistance in racking the bar.
- Keep a grip on the bar until it is racked.



Starting positions



Downward and upward movements

Upward Movement Phase: Spotter

- Keep the hands in the alternated grip position close to—but not touching—the bar as it ascends.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.
- At the athlete's signal after the set is completed, grasp the bar with an alternated grip inside the athlete's hands.
- Guide the bar back onto the supports.
- Keep a grip on the bar until it is racked.

MAJOR MUSCLES INVOLVED

anterior and medial deltoids, triceps brachii

VERBAL CUES

- **Push toward the ceiling**
- **Bar path in front of your face**
- **Push your head through at the top**

COMMON ERRORS

- **Excessive arching of the lower back**
- **Bar path drifting backward**

36 UPRIGHT ROW

Starting Position

- Grasp the bar with a closed, pronated grip approximately shoulder-width apart or slightly wider.
- Stand erect with feet shoulder-width apart, knees slightly flexed.
- Rest the bar on the front of the thighs with the elbows fully extended and pointing out to the sides.

Upward Movement Phase

- Pull the bar up along the abdomen and chest toward the chin.